

Yucheng (Cecilia) Zhang

737-343-4836 | ceciliayczhang@gmail.com | Portfolio | linkedin.com/in/yczcecilia | github.com/yczhangcc

EDUCATION

Northwestern University

M.S. in Computer Science

Aug. 2026 – May 2028

Evanston, IL

The University of Texas at Austin

B.S. in Mathematics, Minor in Computer Science / GPA: 3.74/4.00

Aug. 2023 – May 2026

Austin, TX

Relevant Courses: Data Structure, Stochastic Process, Elements of Software Design, Data Analytics, Database Systems, Discrete Math, Real Analysis, Probability and Statistics, Artificial Intelligence, Elements of Networking

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, R, MATLAB, SAS

Frameworks & Libraries: React, Flask, Node.js, Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Matplotlib, MNE

Databases & Cloud: PostgreSQL, SQL Server, Oracle, Azure Synapse, Databricks, Delta Lake, AWS, Docker

AI/ML & Tools: LLM APIs, OpenAI, Gemini, Claude, LangChain, CrewAI, Git, SQLAlchemy, Alembic, Pytest, Jest, Power BI, REST APIs

WORK EXPERIENCE

Enterprise Risk Analyst Intern

Toyota Industries Commercial Finance, Inc. / Python, SQL Server, Azure, Power BI

May 2025 – Dec. 2025

Dallas, TX

- Developed a full-stack credit risk modeling pipeline from raw SQL Server data, including ETL, feature engineering, Logistic Regression, and Random Forest modeling, improving default prediction accuracy by 3% and reducing manual effort by 10%.
- Validated models with ROC/AUC, confusion matrices, and feature importance, achieving AUC > 0.82 across Class 8 trucks, trailers, and used-vehicle loans and improving risk flagging accuracy by 3–4%.
- Conducted backtesting by comparing predicted risk buckets against observed outcomes, revealing up to 3.7× higher default rates in high-risk groups and informing credit scorecard segmentation refinements.
- Delivered Azure and Power BI dashboards and used advanced Excel reconciliation to align PayNet data with internal loan servicing systems for consistent reporting and real-time risk tracking.

Financial Data Development Engineer – Fraud Detection

Caitong Securities Ltd / SQL, Oracle, Stored Procedures

Jul. 2024 – Aug. 2024

Hangzhou, China

- Designed and optimized SQL-driven ETL pipelines processing 10M+ daily transactions with window functions, subqueries, and stored procedures, improving extraction efficiency by 5%.
- Implemented automated fraud-detection logic with parameterized thresholds and modular stored procedures, reducing false positives by 2% and improving reliability for AML compliance workflows.
- Developed a customer risk scoring engine by integrating multiple transactional and behavioral data streams, enhancing fraud detection precision and financial data monitoring.

Software Engineering Intern – Customer Analytics

Joblogic-X Corporation / Python, APIs, Git

Nov. 2023 – Jun. 2024

Dallas, TX (Hybrid)

- Engineered a scalable data pipeline to scrape and process XHS social-media data and integrate it with transactional order systems for customer analytics.
- Implemented modular ranking and clustering algorithms in Python, generating reproducible customer-priority lists for marketing and engagement workflows.
- Integrated a KOC scoring module into Meet Fresh’s AI-powered chatbot through APIs and maintained Git-based deployment workflows for cross-platform extensibility.

Data Scientist Assistant Intern – Image Restoration

Software Engineering Center, Chinese Academy of Sciences / TensorFlow, Cloud, GFPGAN

Jul. 2023 – Sep. 2023

Beijing, China

- Deployed image restoration modules on Alibaba Cloud via SSH, implementing VAE loss, Gaussian filters, and LSGAN functions for scalable reconstruction workflows.
- Constructed a TensorFlow 2 CNN, partitioned 3.4k images into train/validation/test sets, and achieved 80% accuracy across 5 image categories.
- Integrated GFPGAN APIs for automated face restoration and color correction, reducing processing costs by 5% and supporting model-comparison reporting.

RESEARCH EXPERIENCE AND PROJECTS

CodeAssist Project | GitHub

University of Texas at Austin / Mentor: Prof. Kia Teymourian / React, Flask, PostgreSQL, Docker

Aug. 2025 – Present

Austin, TX

- Built assignment-level AI feedback controls, enabling instructors to configure prompt templates, allowed AI inputs, model settings, request limits, and cooldown rules per assignment.
- Developed Flask REST APIs for AI feedback settings with backend validation, JSON serialization, SQLAlchemy models, Alembic migrations, and instructor/TA authorization safeguards.
- Implemented LLM context-filtering logic so only instructor-approved assignment descriptions, student code, test results, test cases, and student output are sent to AI providers.
- Improved multi-provider AI workflows across OpenAI, Gemini, Claude, and local models, including model selection, API configuration, timeout handling, and settings consistency fixes.

Researcher & TA – Biological Data Analytics

Jan. 2024 – Dec. 2024

University of Texas at Austin | Mentor: Prof. Becca Young | R, Linux, AWS, Docker

Austin, TX

- Processed and analyzed 250GB of RNA-seq data using Linux, Docker, and AWS workflows, improving processing speed by 6% and reducing storage costs by 5%.
- Applied Salmon, RSEM, Trinity, and samtools for transcript quantification, achieving a 98.7% mapping rate and ensuring reproducibility across datasets.
- Performed Voom/Limma differential-expression analysis, identified 48 key genes across time-series samples, and automated normalization, PCA, and volcano-plot workflows in R/ggplot2.
- Collaborated as a teaching assistant to train incoming FRI students on reproducible RNA-seq pipelines and data literacy.

FinChat – AI-Powered Financial Analysis System

Jan. 2023 – May 2024

Southern Methodist University | Mentor: Prof. Labiba Jahan | Python, FinGPT, LangChain, CrewAI

Remote

- Built LLM workflows for financial report analysis by extracting and processing 100+ SEC filings from EDGAR into a scalable data lake for metric extraction and trend detection.
- Implemented LangChain/CrewAI workflows for contextual financial interpretation, improving data processing accuracy by 3% and reducing manual analysis time.

Kiwifruit Quality Grading Project – Machine Learning Member

Aug. 2022 – May 2024

Huazhong Agricultural University | Mentor: Prof. Yan Tong | Python, Computer Vision

Remote

- Contributed to an automated fruit grading pipeline using color, texture, shape, and feature-fusion methods; benchmarked models and analyzed results for the ML methodology section.
- Explored distributed system and blockchain integration for kiwifruit traceability, enhancing data reliability and transparency; work published in *Expert Systems with Applications*.

Machine Learning Research Assistant – Neuroengineering Lab

Jan. 2023 – May 2024

Remote | Mentor: Dr. Kunqiang Qing | Python, MATLAB, MNE

Remote

- Processed 35GB of multimodal EEG/fNIRS data for signal cleaning, visualization, feature extraction, connectivity analysis, and statistical testing.
- Evaluated cognitive patterns and emotion-recognition models using RF, SVM, KNN, and XGBoost, contributing to peer-reviewed work in *Frontiers in Neurology*, *Frontiers in Neuroscience*, and *IEEE ICIPCA*.

LEADERSHIP, COMMUNITY, AND PUBLICATIONS

Leader – Austin Chinese Campus Christian Fellowship

Aug. 2023 – Jun. 2026

University of Texas at Austin

Austin, TX

- Organized weekly Bible studies, retreats, hiking trips, and cultural celebrations; coordinated transportation and scheduling to improve group communication and inclusion.

Volunteer – Art From the Streets

Jan. 2026 – Present

Community art programming for artists experiencing homelessness or housing insecurity

Austin, TX

- Supported event preparation, artwork organization, and visitor engagement, strengthening service-oriented communication and community outreach.

- 1 Miao, H., Wang, Z., **Zhang, Y.**, ... & Tong, Y. (2025). New kiwifruit quality grading methods based on image multi-feature fusion and ResTNet model. *Expert Systems with Applications*, 129507.
- 2 Xu, M., **Zhang, Y.**, Zhang, Y., Liu, X., & Qing, K. (2024). EEG biomarkers analysis in different cognitive impairment after stroke. *Frontiers in Neurology*, 15, 1358167.
- 3 **Zhang, Y.**, Zhang, Y., Jiang, Z., Xu, M., & Qing, K. (2023). The effect of EEG and fNIRS in digital therapy of Alzheimer's disease. *Frontiers in Neuroscience*, 17, 1269359.
- 4 **Zhang, Y.** (2023). EEG-Based on Emotion Recognition Using Machine Learning. *IEEE ICIPCA*, 226–232.
- 5 Li, D., Zhang, Z., & **Zhang, Y.** (2023). Public compassion under the mediation of suffering. *Media and Communication Research*, 4(5), 46–58.